

CS 6650 – Assignment 3 Report

Overview

Implemented a replicated distributed service using a client-driven primary-backup approach to replicate a state-machine log of entries over multiple factory servers. Each server maintains: (i) a thread-safe in-memory map of customer records and (ii) a thread-safe State Machine Log.

Design Summary

- **Roles.** Engineer thread handles customer requests
- **State machine.** Updates the record map (create or set). The log and map are protected by mutexes.
- **Protocol.** Primary appends to log, then sequentially sends information to each backup. After all live backups ack, the primary applies its new updates and advances.
- **Failure handling.** Clients exit cleanly on send/recv errors. Backups resets if the connection dies. Primary ignores failed-backup sockets for commit (treats them as down) and continues with remaining live backups.

What Works / What Doesn't

- **Works:** Single-primary replication; concurrent reads from any server; switching primaries by directing new clients; termination on server/client failures;
- **Not Implemented (Bonus):** Automated recovery & catch-up of repaired servers.

How to Build & Run

- **Build:** `make clean && make.`
- **1 server:** `./server 12345 0 0`
- **3 servers:** `./server 12345 0 2 1 127.0.0.1 22345 2 127.0.0.1 32345, etc.`
- **Client identify:** `./client 127.0.0.1 12345 128 4 1 etc.`

Performance

Write Performance (8 customers, type 1)

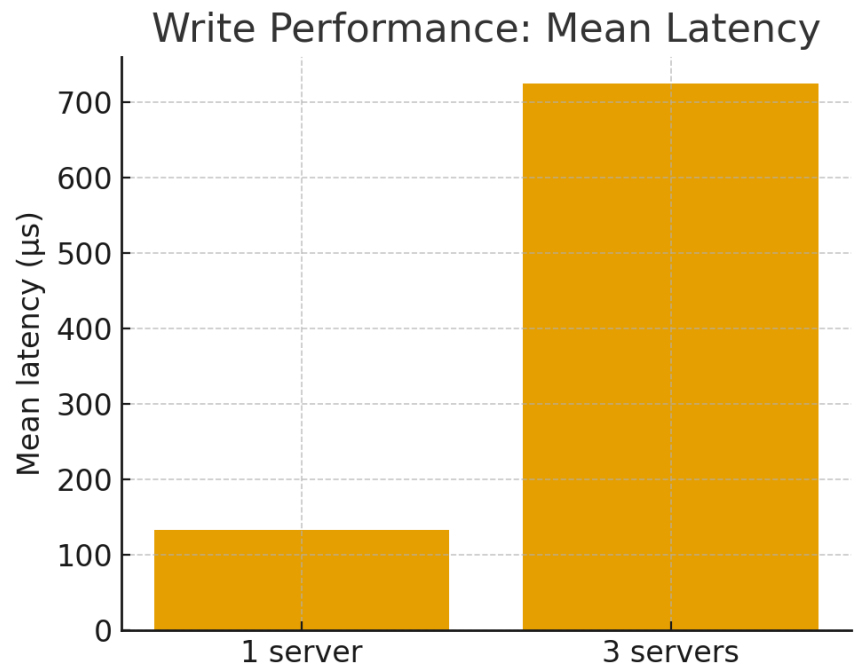


Figure 1: Mean latency for writes.

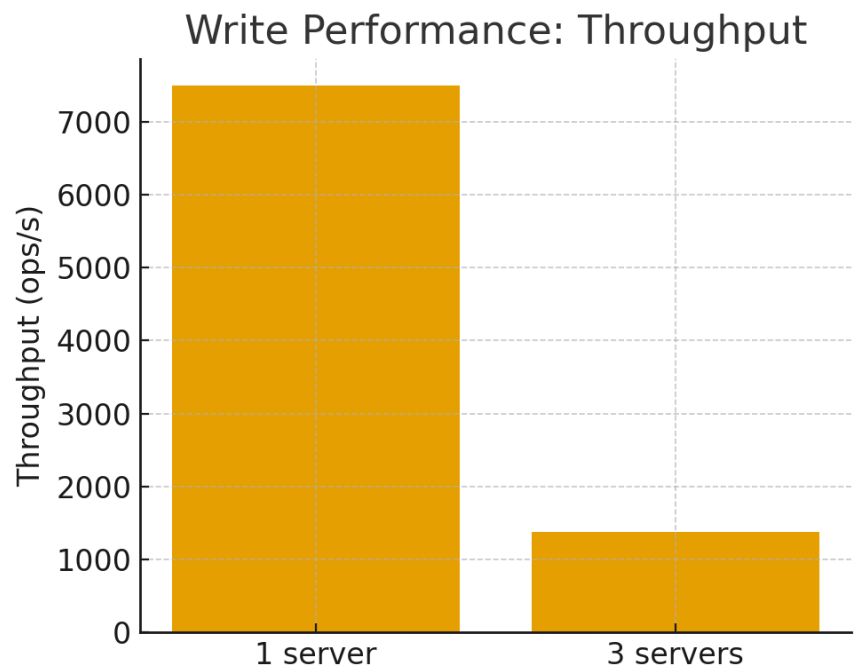


Figure 2: Throughput for writes.

Read Performance (128 customers, type 2)

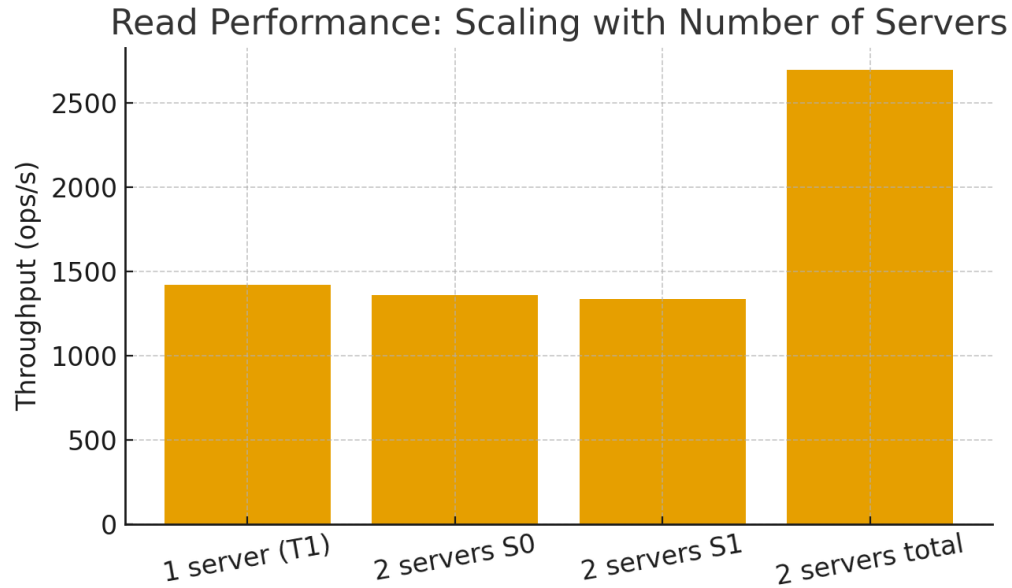


Figure 3: Read throughput scaling with two servers (sum of two concurrent clients).

Raw Measurements

Test	Mean μs	Min μs	Max μs	Throughput (ops/s)
(1 server, write)	133.562	22.233	1165.353	7487.178
(3 servers, write)	723.789	130.374	8993.676	1381.618
Populate - A	1987.703	21.828	11333.800	503.093
Read (1 server)	705.027	12.218	183092.956	1418.385
Populate - B	3639.833	150.290	15877.117	274.738
Read (S0)	735.744	12.813	139636.922	1359.168
2B read (S1)	749.718	11.723	99762.544	1333.835
Total (S0+S1)	—	—	—	2693.003

Replication with failure consideration part

Ran an 8-thread, long-running writer against a 3-node cluster; killed a backup and then the primary. Clients exited cleanly on primary death. Directing a new write to the surviving server promoted it to primary and subsequent scan showed consistent records.

Limitations

The bonus was not implemented. Thanks.